

Integrated Weed Management in Irrigated Rice Fields of Middle Awash, Ethiopia

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Abstract: Weed competition poses a significant challenge to rice productivity, necessitating effective weed management strategies. This study, conducted at the Werer Agricultural Research Center from 2011 to 2013, aimed to determine the most effective weed management practice for irrigated rice fields in Middle Awash, Ethiopia. The research employed a randomized complete block design with various treatments, including mechanical, cultural and chemical control methods: pre-irrigated + Glyphosate 48% EC, pre-irrigated one-hand weeding, pre-irrigated + two-hand weeding, pre-irrigated + three hand weeding, pyoxsylam 45g/L, pyoxsylam 45g/L + one-hand weeding, pyoxsylam 45g/L + two hand weeding, pyoxsylam 45g/L + three hand weeding, Glyphosate 48% EC and pyoxsylam 45g/L, one hand weeding, two hand weeding, three hand weeding and weed free and weedy check. Results indicated that weed management practices significantly ($P < 0.05$) influenced weed density, control efficiency, weed index and grain yield. Integrated management of pre-irrigated rice supplemented with two-hand weeding demonstrated superior weed control efficiency, reducing yield loss by 88%. These findings underscore the importance of integrated weed management practices in mitigating weed competition and enhancing rice production in irrigated fields. Further research is recommended to explore additional strategies for effective weed control in rice cultivation.

Key words: Weed Flora • Weed Density • Weed Index • Weed Control Efficiency • Yield

INTRODUCTION

Rice (*Oryza sativa* L.) is an important cereal crop and the preferred staple food of more than half of the world's population [1]. The production of crops was 518.5 million tons in 2020-22 [2]. Sub-Saharan Africa (SSA) accounts for 6% and 26% of global total rice production and harvested area, respectively [3]. In Ethiopia, rice production and area coverage were 189, 649 tons and 62, 551 hectares, respectively in 2020 [3]. Ethiopian rice yield (3.34 t/ha) was higher than both African and East African productivity, but lower than the global average [3].

Direct-seeded rice is gaining popularity due to labor shortages during the peak transplanting season and the availability of water for short periods [4]. Rice planted directly needs only 34% of the total labor required and costs 27% less than a transplanted crop overall [5]. Additionally, the weed competition is greater in direct-seeded rice than in transplanted rice because both weed

and crop seeds emerge at the same time and compete with each other from the germination, resulting in lower grain yield.

Weeds are a significant biological barrier to limiting crop growth and productivity [6]. It reduces the yield by 50-60% in transplanted rice and 70-80% in direct-seeded rice [7], in some cases, crop failure has been observed [8]. Crop-weed competition decreases rice yield by 40-60%, while it can rise to 94-96% for season-long competition [9-11]. This yield loss can be mitigated by effective and integrated weed management practices. There are various weed control strategies, some of which have been proven beneficial, some of which are cost-effective and others that save time [12].

Weed management in rice is done manually and chemically. Manual weeding is a very common and effective but labor-intensive method [13]. Chemical weed control is the most widely used approach in rice because it is simple, fast, time-saving, affordable and reliable [14].

Integration of hand weeding with different pre-emergence or post-emergence herbicides amplifies their efficacy [15, 16]. To achieve food self-sufficiency by producing food crops widely in the country. Finding a sustainable weed management option is very important. In irrigated lowland, Werer areas, weed infestation is high in rice fields. However, no study has been done on weed control options in rice and, none of the weeds known to infest rice in this area have been identified. Therefore, this study aims to determine the effective weed management practice that can be used in irrigated rice.

MATERIALS AND METHODS

The field experiment was conducted at the Werer Agricultural Research Center during the main cropping season from 2011 to 2013. The rice variety NERICA- 4 was sown on plots of 1.8m x 5m (9m²). A randomized complete block design with three replications was used. The treatments were comprised of 14 variants: pre-irrigated (before sowing) + Glyphosate 48% EC @ 2L/ha, pre-irrigated + one-hand weeding, pre-irrigated + two-hand weeding, pre-irrigated + three-hand weeding, pyoxsylam 45g/L @ 450ml/ha, pyoxsylam 45g/L + one-hand weeding, pyoxsylam 45g/L @ 450ml/ha + two-hand weeding, pyoxsylam 45g/L @ 450ml/ha + three-hand weeding, Glyphosate 48% EC @ 2L/ha and pyoxsylam 45g/L @ 450ml/ha, one hand weeding only, two hands weeding only, three hands weeding only and weed free and weedy check. Hand weeding was done for each treatment at 2, 4 and 6 weeks after rice crop emergence depending on the number of weedings applied. Data on stand count at the seedling stage, type and the number of weeds that emerged at the seedling stage to crop maturity (1m² quadrant/plot counts before and after spray) and yield data were recorded. All other recommended agronomic practices were done.

Data Analysis: Weed Relative Density Weed Control Efficiency (WCE) and Weed Index for all treatments were calculated using the following formula [17]

$$\text{Relative Density(\%)} = \frac{\text{Number of individuals of target species occurred}}{\text{Number of individuals of all the species occurred}} * 100$$

$$\text{Weed Control Efficiency (\%)} = \frac{\text{WC-WT}}{\text{WC}} * 100$$

where,

WC - the weed density (no. m⁻²) in the control plot;

WT - the weed density (no. m⁻²) in the treatment plot

$$\text{Weed Index (\%)} = \frac{\text{Yield from weed free plot} - \text{Yield from a treated plot}}{\text{Yield from a weed-free plot}} * 100$$

Finally, data on weed density, weed control efficiency, weed index and yield data were analyzed by SAS statistical analysis package 9.4 and ANOVA to test the significant differences between treatments.

RESULTS

Weed Density: The result showed 33 weed species belonging to 19 families, which mainly included Gramineae, Fabaceae, Asteraceae and Convolvulaceae families were recorded (Table 1). The dominant weeds in the area were *Cyperus* spp among sedge, *Eriocloa fatmensis* and *Corchorus* spp among grass and broadleaved, respectively (Table 1).

The weed density was significantly affected by weed management practices (Table 2). Pre-irrigation + three-hand weeding had the lowest weed density and weedy check had the most in all cropping seasons (Table 2). Similarly, the overall data analysis shows that pre-irrigation + three-hand weeding significantly reduced the weed density in the study area (Table 2). Weed infestation was higher in the 2011 cropping season than in 2012 and 2013. This could be attributed to the high rainfall in the 2011 cropping season.

Weed Control Efficiency and Weed Index: Weed control efficiency was significantly affected by the management practices (Table 3). The highest weed control efficiency was recorded by pre-irrigated + 3-hand weeding (57%) while the lowest weed control efficiency was obtained from one-hand weeding.

Weed management practices have a significant impact on weed control efficiency and the weed index as well (Table 3). Pre-irrigated + 3 hand weeding had the highest weed control efficacy, which was statistically comparable to pre-irrigated + two hand weeding and pre-irrigated + one hand weeding. While. the weedy check had the lowest weed control efficiency, followed by once-hand weeding (Table 3). The highest weed index was found in Pyoxsylam 45g/L + one-hand weeding plots (94.44), followed by Pyoxsylam 45g/L (66.36) plots and the lowest weed index was found in pre-irrigated + three-hand weeding plots. The weed index of the weed-free plot differed significantly from all other management practices (Table 3).

Effect of Weed Management on Yield of Rice: The yield of treatments significantly differed showing the effect of

Table 1: Weed species population and Relative weed density

No	Scientific name	Family	Life type	2011		2012		2013	
				PP (M ²)	RWD (%)	PP (M ²)	RWD (%)	PP (M ²)	RWD (%)
1	<i>Cyperus spp.</i>	Cyperaceae	Sedge	52.65	21.00	24.08	30.22	38.37	32.25
2	<i>Eriocloa fatmensis</i>	Gramineae	Grass	29.68	11.84	15.39	19.32	22.54	18.94
3	<i>Corchorus spp.</i>	Tiliaceae	Broad	25.85	10.31	11.56	14.51	18.70	15.72
4	<i>Setaria pumila</i>	Gramineae	Grass	20.27	8.09	5.99	7.51	13.13	11.04
5	<i>Echinochloa colona (L.)</i>	Gramineae	Grass	15.78	6.29	1.49	1.87	8.63	7.26
6	<i>Portulaca oleracea L.</i>	portulacaceae	Broad	11.88	4.74	3.31	4.16	4.74	3.98
7	<i>Acalypha crenata</i>	Euphorbiaceae	Broad	11.60	4.63	3.03	3.80	4.45	3.74
8	<i>Sorghum arundinaceum</i>	Gramineae	Grass	10.57	4.21	2.71	3.40	3.42	2.88
9	<i>Barleria ventricosa</i>	Acanthaceae	Broad	8.44	3.37	1.30	1.63	1.30	1.09
10	<i>Launaea cornuta</i>	compositae	Broad	8.58	3.42	8.58	10.76	1.43	1.20
11	<i>Desmodium spp.</i>	Fabaceae	Broad	7.86	3.14	0.72	0.90	0.72	0.60
12	<i>Portulaca oleracea L.</i>	Asteraceae	Broad	5.89	2.35	0.18	0.22	0.18	0.15
13	<i>Ipomea eriocarpa R. Br.</i>	Convolvulaceae	Broad	5.84	2.33	0.12	0.15	0.12	0.10
14	<i>Polypogon monspeliensis</i>	Gramineae	Broad	5.51	2.20	0.51	0.64	0.51	0.43
15	<i>Amaranthus hybridus L.</i>	Amaranthaceae	Broad	5.14	2.05	0.14	0.17	0.14	0.11
16	<i>Commelina benghalensis L.</i>	Commelinaceae	Broad	5.01	2.00	0.01	0.01	0.01	0.01
17	<i>Prosopis juliflora (Sw.) DC.</i>	Fabaceae	Tree weed	4.34	1.73	0.06	0.07	0.06	0.05
18	<i>Boerhaavia erecta L.</i>	Nyctaginaceae	Broad	3.68	1.47	0.11	0.14	0.11	0.09
19	<i>Ipomoea abyssinica</i>	Convolvulaceae	Broad	3.65	1.46	0.08	0.10	0.08	0.07
20	<i>Phyllanthus spp</i>	Phyllanthaceae	Broad	2.33	0.93	0.19	0.23	0.19	0.16
21	<i>Ipomoea mauritiana</i>	Convolvulaceae	Broad	2.15	0.86	0.00	0.00	0.00	0.00
22	<i>Tribulus terrestris L.</i>	Zygophyllaceae	Broad	0.50	0.20	0.00	0.01	0.00	0.00
23	<i>Chenopodium ambrosioides L.</i>	Chenopodiaceae	Broad	0.48	0.19	0.05	0.06	0.05	0.04
24	<i>Chenopodium procerum</i>	Chenopodiaceae	Broad	0.43	0.17	0.00	0.00	0.00	0.00
25	<i>Chenopodium album L.</i>	Amaranthaceae	Broad	0.44	0.17	0.01	0.01	0.01	0.01
26	<i>Zaleya pentandra (L.)</i>	Aizoaceae	Broad	0.38	0.15	0.00	0.00	0.00	0.00
27	<i>Xanthium strumarium L.</i>	Asteraceae	Broad	0.36	0.14	0.00	0.00	0.00	0.00
28	<i>Crotalaria spp</i>	Fabaceae	Broad	0.36	0.14	0.00	0.00	0.00	0.00
29	<i>Gynandropsis gynandra (L.)</i>	capparideae	Broad	0.32	0.13	0.04	0.05	0.04	0.03
30	<i>Datura stramonium L.</i>	solanaceae	Broad	0.25	0.10	0.03	0.04	0.03	0.03
31	<i>Boerhavia coccinea</i>	Nyctaginaceae	Broad	0.23	0.09	0.00	0.00	0.00	0.00
32	<i>Phalaris paradoxa L.</i>	Gramineae	Grass	0.22	0.09	0.00	0.00	0.00	0.00
33	<i>Guizotia scabra (Vis.)</i>	Asteraceae	Broad	0.08	0.03	0.00	0.01	0.00	0.00

Note: PP= plant population, RWD=Relative weed density

Table 2: Effect of weed management on weed density

		2011 WD	2012 WD	2013 WD	Mean
1	Pre-irrigated + Glyphosate 48% EC	235.9 ^{bc}	99.3 ^{ab}	135.9 ^{bc}	157 ^{bc}
2	Pre-irrigated +1 hand weeding	239 ^{bc}	89.0 ^{ab}	139 ^{bc}	155.7 ^{bc}
3	Pre-irrigated + 2 hand weeding	186.9 ^{cde}	40.3 ^{cde}	86.9 ^{cde}	104.7 ^{cde}
4	Pre-irrigated + 3 hand weeding	168.9 ^{de}	38.9 ^{cde}	68.9 ^{de}	92.3 ^{de}
5	Pyoxsylum 45g/L	220.9 ^{bcd}	78.9 ^{bcd}	120.9 ^{bc}	140.3 ^{bcd}
6	Pyoxsylum 45g/L +1 hand weeding	229.8 ^{bcd}	83.1 ^{bcd}	129.8 ^{bc}	147.5 ^{bcd}
7	Pyoxsylum 45g/L + 2 hand weeding	216.9 ^{bcd}	73.5 ^{bcd}	116.9 ^{bc}	135.8 ^{bcd}
8	Pyoxsylum 45g/L + 3 hand weeding	236.9 ^{bc}	86.9 ^{bc}	136.9 ^{bc}	153.6 ^{bc}
9	Glyphosate 48% EC and Pyoxsylum 45g/L	242.6 ^{bc}	82.5 ^{bcd}	142.6 ^{bc}	155.9 ^{bc}
10	Once hand weeding	257 ^{ab}	110.3 ^{ab}	157 ^{ab}	174.8 ^{ab}
11	Twice hand weeding	236.5 ^{bc}	83.1 ^{bcd}	136.5 ^{bc}	152.0 ^{bc}
12	Three-hand weeding	186.8 ^{cde}	36.8 ^{de}	86.8 ^{cde}	103.5 ^{cde}
13	Weed free	151.5 ^d	28.2 ^d	51.5 ^e	77.1 ^e
14	Weedy check.	313.1 ^a	136.5 ^a	213.1 ^a	220.9 ^a
	Mean	223.05	76.25	123.06	140.79
	C.V. (%)	17.39	27.73	31.53	24.17
	LSD (0.05%)	65.128	48.278	65.128	57.09

Table 3: Effect of weed management on weed control efficiency and weed index

		WCE (%)	WI (%)
1	Pre-irrigated + Glyphosate 48% EC	25.8 ^e	71.39 ^{ab}
2	Pre-irrigated +1 hand weeding	52.2 ^{abc}	60.09 ^{bcd}
3	Pre-irrigated + 2 hand weeding	51.9 ^{abcd}	55.56 ^{bcd}
4	Pre-irrigated + 3 hand weeding	57.21 ^{ab}	28.21 ^{de}
5	Pyoxsylam 45g/L	36.18 ^{bcd}	66.36 ^{abc}
6	Pyoxsylam 45g/L +1 hand weeding	29.29 ^{de}	94.44 ^a
7	Pyoxsylam 45g/L + 2 hand weeding	36.1 ^{bcd}	66.27 ^{abc}
8	Pyoxsylam 45g/L + 3 hand weeding	27.2 ^{de}	53.63 ^{cd}
9	Glyphosate 48% EC and Pyoxsylam 45g/L	31.4 ^{cde}	51.65 ^{cd}
10	Once hand weeding	18.7 ^{ef}	64.67 ^{abc}
11	Twice hand weeding	28.03 ^{cde}	62.67 ^{abc}
12	Three-hand weeding	30.95 ^{cde}	57.75 ^{bcd}
13	Weed free	63.88 ^a	0 ^e
14	Weedy check.	0 ^f	87.66 ^{ab}
	Mean	34.91	58.59
	C.V. (%)	22.45	24.16
	LSD (0.05%)	24.87	33.597

Table 4: Mean effect of integrated weed management on yield of rice (2011-2013)

No	Treatments	Yield (ton/ha)			
		2011	2012	2013	Mean
1	Pre-irrigated + Glyphosate 48% EC	0.17 ^{cdef}	0.657 ^d	0.639 ^{bc}	0.489 ^{defg}
2	Pre-irrigated +1 hand weeding	0.178 ^{cdef}	1.473 ^{bc}	1.017 ^{abc}	0.889 ^{abc}
3	Pre-irrigated + 2 hand weeding	0.467 ^a	1.237 ^c	1.814 ^a	1.173 ^a
4	Pre-irrigated + 3 hand weeding	0.496 ^a	1.887 ^a	1.061 ^{abc}	1.19 ^a
5	Pyoxsylam 45g/L	0.183 ^{cde}	0.5 ^d	0.831 ^{abc}	0.505 ^{defg}
6	Pyoxsylam 45g/L +1 hand weeding	0.304 ^b	0.13 ^{efg}	0.086 ^c	0.173 ^{fg}
7	Pyoxsylam 45g/L + 2 hand weeding	0.264 ^{bc}	0.463 ^{de}	0.903 ^{abc}	0.543 ^{cdef}
8	Pyoxsylam 45g/L + 3 hand weeding	0.134 ^{def}	0.367 ^{def}	1.239 ^{ab}	0.579 ^{bcd}
9	Glyphosate 48% EC and Pyoxsylam 45g/L	0.113 ^{ef}	0.57 ^d	1.094 ^{abc}	0.592 ^{bcd}
10	Once hand weeding	0.074 ^{ef}	0.053 ^{fg}	0.906 ^{abc}	0.344 ^{efg}
11	Twice hand weeding	0.072 ^{ef}	1.54 ^{bc}	0.911 ^{abc}	0.841 ^{abcd}
12	Three-hand weeding	0.077 ^{ef}	1.723 ^b	0.978 ^{abc}	0.926 ^{ab}
13	Weed free	0.243 ^{bcd}	1.47 ^{bc}	1.858 ^a	1.215 ^a
14	Weedy check.	0.065 ^f	0 ^g	0.345 ^{bc}	0.137 ^g
	Mean	0.203	0.876	0.977	0.685
	CV %	24.472	23.783	23.834	26.028
	LSD (0.05%)	0.117	0.349	1.047	0.379

weed management practices (Table 4). The maximum yield was obtained from weed free followed by pre-irrigated + three-hand weeding treatment, followed by weed-free and pre-irrigated + two-hand weeding (Table 4). The pre-irrigated + three-hand weeding treatment had the highest yield, followed by weed-free and pre-irrigated + two-hand weeding, which could be because of dry time weed management that causes the damage of rice seedlings during frequent weeding to make the plot weed-free (Table 4).

DISCUSSION

The data showed the experimental field was infested with a mixed weed flora of sedges, grasses and broad-

leaved. Among weeds, the Cyperaceae family had the highest species richness and density compared to other weeds. This could be a result of having a large number of seeds of sedges stored from the past years in the soil seed bank. The results are consistent with the findings of Kamoshita *et al.* [18] Uddin *et al.* [19] and Aravindh *et al.* [20] who reported for many species in this family seed bank development is a key component of regeneration. A similar result also obtained on rice fields was infested by grasses, broad leave and sedge weeds [21- 24].

Our study showed integrated management of pre-irrigated with three-hand weeding had the lowest weed density, weed index and the highest weed control efficiency compared with other weed management practices. This indicates the relative effectiveness of

these weed management options in weed controlling and decreasing yield loss in irrigated rice of the Middle Awash of Ethiopia. Our finding correlated with Kumar *et al.* [25] and Badal Verma *et al.* [26] who found Weed management practices significantly reduced weed density when compared with weedy check plots

The data showed that pre-irrigated supplemented with two or three-hand weeding reduces a yield loss by 88.37% and 88.7% respectively. The weed-free treatment was the best yielder and pre-irrigated with three-hand weeding was the next high yielder. This was mostly caused by a decrease in weed competition in the plots where these treatments were used. While the lowest is from the weedy check. This is mainly due to there being high weed competition present in the plots which resulted in the lowest yield. Our finding is in line with Priya *et al.* [22] and Sachin [27] who stated that weed-free had the maximum yielder, followed by two-hand weeding and hoeing. Many scholars reported weed control through pre-irrigated with two or three hand weeding reduced the weed population and maximized the yield in the rice field [22, 28-30].

CONCLUSION

The study highlights the significant impact of weed management practices on weed density, control efficiency, weed index and grain yield in irrigated rice fields in Middle Awash, Ethiopia. The study indicates rice fields are plagued with a diverse weed flora and *Corchorus spp.*, *Eriocloa fatmensis* and *Cyperus spp* are the three most prevalent weeds in the areas. The results demonstrate that integrated management of pre-irrigated rice supplemented with two-hand weeding is an effective strategy for improving weed control efficiency and reducing yield loss by 88%. This integrated approach shows promise in mitigating weed competition and enhancing rice productivity in irrigated fields. The findings underscore the importance of adopting integrated weed management practices to optimize rice production and minimize the negative effects of weed competition. Further research is recommended to explore additional weed management strategies and assess their economic implications for rice farmers in similar environmental conditions. Overall, the study contributes valuable insights into effective weed management practices for irrigated rice cultivation and emphasizes the need for integrated approaches to address weed challenges in rice fields.

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